Introduction to Field Programmable Gate Arrays

Laboratory Assignment 6
ECE 201: Digital Circuits and Systems

Learning Objectives

After completing this lab, you will be able to

- explain the basics of field programmable gate arrays
- describe logic circuits using a hardware description language and synthesize them on to an FPGA chip
- connect simple input and output devices to an FPGA chip

Equipment and Materials

- DE10-Lite Board
- Intel Quartus Prime Design Software v15.1 or v18.1

Pre-Lab Assignment

Prior to attending your assigned laboratory section, complete the following tasks

- Browse through the datasheet for the FPGA you will use, the Intel Max 10 FPGA. Specifically, look at the chip floorplan shown in **Figure 1** and the programmable logic elements shown in **Figure 5**.
- Watch just the opening segment of this video (stop at the 4:50 mark) Basics of Programmable Logic: FPGA Architecture Starting around the 3:30 mark, the narrator explains how Lookup Tables are implemented in the FPGA you are using.
- Familiarize yourself with the user manual for the FPGA development board that houses your FPGA chip, the DE10-Lite.
- Review the VHDL code given in lecture.

Part I: Tutorials

Follow the tutorials linked on BB to become familiar with Quartus, ModelSim, and programming the FPGA.

Part II: Rock-Paper-Scissors!

In this lab, you will see how your design of Rock-Paper-Scissors from Lab 5 can be implemented in VHDL on an FPGA. The DE10-Lite board provide ten switches and lights, called SW_{9-0} and $LEDR_{9-0}$. The switches can be used to provide inputs, and the lights can be used as output devices. Four switches on the board will be used to provide the inputs A_1, A_0, B_1, and B_0. Three LEDs will be used for W_A, E, and W_B.

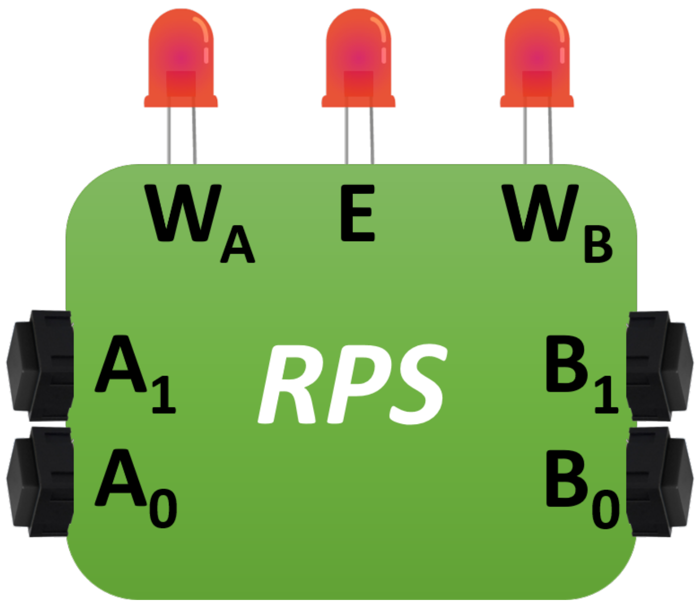

Figure 1: Blackbox view of the Rock-Paper-Scissors Game

The DE-series boards have hardwired connections between its FPGA chip and the switches and lights. To use the switches and lights, it is necessary to include in your Quartus project the correct pin assignments, which are given in your board's user manual.

Perform the following steps to implement you Lab 5 Rock-Paper-Scissors circuit on your FPGA:

1. Create a new Quartus project for your circuit. Select the target chip that corresponds to your DE-series board (10M50DAF484C7G).
2. Create a VHDL file containing the starter code in Figure 2 and include it in your project. Copying and pasting VHDL from this PDF is likely to cause issues. In order to avoid these issues and to help you remember the syntax, please retype it manually.

```
LIBRARY ieee;
USE ieee.std_logic_1164.all;

ENTITY RPS IS
   PORT (
            A    : IN    STD_LOGIC_VECTOR(1 DOWNTO 0);
            B    : IN    STD_LOGIC_VECTOR(1 DOWNTO 0);
            WA   : OUT   STD_LOGIC;
            WB   : OUT   STD_LOGIC;
            E    : OUT   STD_LOGIC
   );
END RPS;

ARCHITECTURE comb OF RPS IS
BEGIN
   . . .
END comb;
```

Figure 2: Top level VHDL code for Rock-Paper-Scissors.

3. Program your circuit from Lab 5 using VHDL. That is, code the architecture that is missing from the Figure 2 starter code. **Use only signal assignments and logic expressions**, that is, do not use "when/else" or similar VHDL statements. Redraw your circuit in your submission template.
4. Compile your design.
5. Simulate your design using ModelSim. Create a do file that executes all possible input combinations.
6. Use the Pin Planner in Quartus to map pins on the FPGA to the inputs and outputs of your design.
7. Recompile your design.
8. Download the compiled circuit into the FPGA chip by using the Quartus Programmer tool. Test the functionality of the circuit by toggling the switches and observing the LEDs.
9. **Demonstrate the functionality of your FPGA implementation for a lab checkoff.**